

Statistics

2025 Lectures

Part 4 - Expectations

Institute of Economic Studies
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Recommended reading (week 3)

For the week 3 lecture (Part 4), please, read

- sections 7.1, 7.2, 7.4, 7.5, 7.6 from Bartoszynski and Niewiadomska-Bugaj (33 pages)
- sections 3.1, 3.2, 3.3, 3.4 from Mittelhammer (15 pages)

Motivation example for expections

You invest 100\$ in a risky asset:

- with probability 0.6 you receive 120\$
- with probability 0.4 you receive 80\$.

$$E[\text{payoff}] = 0.6 \cdot 120 + 0.4 \cdot 80 = 104$$

$$\Rightarrow E[\text{net return}] = \frac{104 - 100}{100} = 4\%.$$

Motivation for expected value

Example 29: Consider a random variable X defined on a sample space S . An experiment involves randomly selecting a point $s \in S$, and $X(s)$ can be interpreted as the gain (or loss) of a gambler.

If the experiment is repeated n times, the outcome is represented by $s^{(n)} = (s_1, \dots, s_n)$, where $s_i \in S, i = 1, \dots, n$. This leads to the gambler's total accumulated gain $X(s_1) + X(s_2) + \dots + X(s_n)$.

As n increases, the average gain per gamble fluctuates less and less gradually stabilizes at a value known as the expected value of X .

- the expected value of a rv can be understood as the “balance point” of the distribution of probability on the real line

Expectation of discrete rv's

Definition 14: If X is a discrete random variable that assumes values $x_1, x_2, \dots$ with probabilities $P(X = x_i), i = 1, 2, \dots$, then the **expected value** of X is defined as

$$E(X) = \sum_{i=1}^{\infty} x_i P(X = x_i)$$

provided

$$E|X| = \sum_{i=1}^{\infty} |x_i| P(X = x_i) < \infty.$$

Example 30: Let event A occurs with probability p , and you win USD 1 if A occurs. Otherwise you loose USD 1. Calculate the expected value of this game.

Example 31: Let event A occurs with probability p , and you win USD 1 if A occurs. Otherwise you loose USD 0. Calculate the expected value of this game.

Examples

Example 32: (Binomial distribution)

$$E(X) = \sum_{k=0}^n k \cdot \binom{n}{k} p^k (1-p)^{n-k} = np.$$

Example 33: (Geometric distribution)

$$E(X) = \sum_{k=0}^{\infty} k \cdot (1-p)^k p = \frac{1-p}{p}.$$

Example 34: (Petersburg paradox)

A fair coin is tossed repeatedly until the first head appears. If the first head occurs on the k th toss, you win 2^k dollars. How much should you be willing to pay to participate in this game?

You should play for a fee less than the expected value of the game:

$$E(X) = 2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{2^2} + \cdots = +\infty!$$

$EX = \pm\infty$ vs. EX does not exist

If the condition in Definition 14 is not met then we can consider (for the discrete case)

$$U^+ = \sum (\max(x_i, 0)P(X = x_i))$$

$$U^- = - \sum (\min(x_i, 0)P(X = x_i))$$

Then $E|X| = U^+ + U^-$ and

$$|E(X)| < \infty \Leftrightarrow U^+ < \infty, U^- < \infty$$

$$E(X) = +\infty \Leftrightarrow U^+ = \infty, U^- < \infty$$

$$E(X) = -\infty \Leftrightarrow U^+ < \infty, U^- = \infty$$

$$E(X) \text{ does not exist} \Leftrightarrow U^+ = \infty, U^- = \infty$$

Expectation of continuous rv's

Definition 15: If X is a continuous random variable with the density f_X , then the **expected value** of X is defined as

$$E(X) = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$$

provided that

$$E|X| = \int_{-\infty}^{\infty} |x| \cdot f_X(x) dx < \infty.$$

Example 35: (Uniform distribution on $[a, b]$)

$$E(X) = \int_a^b \frac{x}{b-a} dx = \frac{a+b}{2}.$$

More examples

Example 36: (Exponential distribution)

$$E(X) = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}.$$

Example 37: (Normal distribution)

$$\begin{aligned} E(X) &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx = \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + z\sigma) e^{-\frac{z^2}{2}} dz = \\ &= \mu \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz + \frac{\sigma}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz = \\ &= \mu. \end{aligned}$$

Expectation and functions of random variables

Theorem 10: Let X be a random variable. In the discrete case, if g is a real function such that $\sum_i |g(x_i)|P(X = x_i) < \infty$ then

$$E(g(X)) = \sum_i g(x_i)P(X = x_i).$$

In the continuous case if $\int_{-\infty}^{\infty} |g(x)| \cdot f_X(x) dx < \infty$ then

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx.$$

Properties of the expected value

Theorem 11:

a) If $E(X)$ exists then for all $a, b \in \mathbb{R}$

$$E(aX + b) = aE(X) + b.$$

b) If $X \geq 0$ then $E(X) \geq 0$.

c) For any random variable X

$$|E(X)| \leq E|X|.$$

d) If $|X| < Y$ and $E(Y) < \infty$ then also $E(X) < \infty$.

Expectation and functions of random variables

Theorem 12: Let X, Y be random variables. In the discrete case, if $h(\cdot, \cdot)$ is a real function such that $\sum_{i,j} |h(x_i, y_j)| p_{ij} < \infty$ then

$$E(h(X, Y)) = \sum_i \sum_j h(x_i, y_j) p_{ij} = \sum_j \sum_i h(x_i, y_j) p_{ij}.$$

In the continuous case if

$$\iint_{\mathbb{R}^2} |h(x, y)| \cdot f(x, y) \mathbf{d}(x, y) < \infty$$

then

$$\begin{aligned} E(h(X, Y)) &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} h(x, y) \cdot f(x, y) \mathbf{d}x \right) \mathbf{d}y \\ &= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} h(x, y) \cdot f(x, y) \mathbf{d}y \right) \mathbf{d}x. \end{aligned}$$

Expectation and functions of random variables

Theorem 13: Assume that $E|X| < \infty$ and $E|Y| < \infty$. Then

$$E(X + Y) = E(X) + E(Y).$$

Consequently,

$$E\left(\sum_i (a_i X_i) + b\right) = \sum_i (a_i E(X_i)) + b$$

and, applying also Theorem 11 b), if $Y \leq Z$ then $E(Y) \leq E(Z)$.

Theorem 14: If X and Y are independent random variables and g, h real valued functions such that $E|g(X)h(Y)| < \infty$ then

$$E(g(X)h(Y)) = E(g(X)) \cdot E(h(Y)).$$

Moments of random variables

Definition 16: For any random variable X , the expectation of X^n (if exists) is called the n th ordinary moment of X

$$m_n = E(X^n).$$

An absolute moment of order n is defined as

$$\beta_n = E|X^n|.$$

Example 38: (Poisson distribution)

$$m_1 = E(X) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} = \lambda.$$

$$m_2 = E(X^2) = \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} = \lambda^2 + \lambda.$$

Existence of moments and Liapunov inequality

Example 39: A degenerate random variable is a random variable that takes a constant value. That is, $X = c$ with $P(X = c) = 1$.

$$E(X) = c \cdot 1 = c, \quad E(X^2) = c^2 \cdot 1 = c^2.$$

Thus, $m_n = c^n$.

Definition 17: Ordinary moments of $X - E(X)$ are called **central moments** of X

$$\gamma_n = E(X - m_1)^n.$$

Theorem 15: If an absolute moment of order m exists then all moments (ordinary, absolute, central) of orders $n \leq m$ exist.

Theorem 16: (Liapunov inequality)

If $0 < \alpha < \beta < \infty$ then

$$[E(|X|^\alpha)]^{\frac{1}{\alpha}} \leq [E(|X|^\beta)]^{\frac{1}{\beta}}.$$

Moment generating function

Definition 18: The function of real variable t defined as

$$m_X(t) = E(e^{tX}), t \in \mathbb{R},$$

is called a **moment generating function** (mgf).

- For any random variable, its mgf exists for $t = 0$.

Example 40: (Poisson distribution)

$$m_X(t) = \sum_{k=0}^{\infty} e^{tk} \frac{\lambda^k}{k!} e^{-\lambda} = e^{\lambda(e^t-1)}, \quad t \in \mathbb{R}.$$

Example 41: (Exponential distribution)

$$m_X(t) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \frac{\lambda}{\lambda - t}, \quad \text{if } t < \lambda.$$

Moments and moment generating function

Theorem 17: Let $m_X(t) < \infty$, $|t| < t_0$ for some $t_0 > 0$. Then $E|X|^r < \infty$ for all $r > 0$ and

$$m_n = \left. \frac{\partial^n m_X(t)}{\partial t^n} \right|_{t=0} \left(= m_X^{(n)}(0) \right).$$

Example 42: (Binomial distribution)

$$m_X(t) = \sum_{k=0}^n \left[e^{tk} \binom{n}{k} p^k q^{n-k} \right] = (pe^t + q)^n, \quad t \in \mathbb{R}.$$

We also have

$$\begin{aligned} m_X'(t) &= n(pe^t + q)^{n-1} pe^t, \\ m_X''(t) &= n(n-1)(pe^t + q)^{n-2} (pe^t)^2 + n(pe^t + q)^{n-1} pe^t. \end{aligned}$$

Thus $E(X) = m_X'(0) = np$ and $E(X^2) = n(n-1)p^2 + np$.

Properties of mgf's

Theorem 18: If X is a random variable with $m_X(t)$ and $a, b \in \mathbb{R}$ then $Y = aX + b$ has mgf

$$m_Y(t) = e^{bt} m_X(at), \quad t \in \mathbb{R}.$$

Example 43: $Z \sim N(0, 1), X = \mu + \sigma Z, X \sim N(\mu, \sigma^2)$

$$m_Z(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{tz} e^{-\frac{z^2}{2}} dz = e^{\frac{t^2}{2}}, \quad t \in \mathbb{R},$$

$$m_X(t) = e^{\mu t + \frac{\sigma^2 t^2}{2}}, \quad t \in \mathbb{R}.$$

Theorem 19: If X and Y are independent random variables with moment generating functions $m_X(t), t \in \mathbb{R}$ and $m_Y(t), t \in \mathbb{R}$ then mgf of $X + Y$ is

$$m_{X+Y}(t) = m_X(t) \cdot m_Y(t), \quad t \in \mathbb{R}.$$

Properties of mgf's

Theorem 20: If X and Y are two random variables such that their moment generating functions $m_X(t), t \in \mathbb{R}$, and $m_Y(t), t \in \mathbb{R}$, coincide in some neighborhood of $t = 0$, then X and Y have the same distribution.

Example 44: Let X and Y be independent random variables, each following a Poisson distribution with means λ_1 and λ_2 , respectively. Then, their mgfs are

$$m_X(t) = e^{\lambda_1(e^t-1)}, t \in \mathbb{R}, \quad m_Y(t) = e^{\lambda_2(e^t-1)}, t \in \mathbb{R}.$$

By Theorem 19, the mgf of $X + Y$ is

$$m_{X+Y}(t) = m_X(t)m_Y(t) = e^{(\lambda_1+\lambda_2)(e^t-1)}, t \in \mathbb{R},$$

which corresponds to the mgf of a Poisson distribution with mean $\lambda_1 + \lambda_2$. By Theorem 20, we conclude that $X + Y \sim POI(\lambda_1 + \lambda_2)$.

Variance and standard deviation

Definition 19: If $E(X^2) < \infty$ then the second central moment of X

$$\text{Var}X = E[(X - E(X))^2]$$

is called the **variance** of X . Its positive square root is called the **standard deviation** of X , often denoted by σ .

- since it is expectation of $(X - E(X))^2 \geq 0$, variance is always nonnegative



$$\text{Var}X = E(X^2 - 2X \cdot E(X) + (E(X))^2) = E(X^2) - (E(X))^2.$$

Theorem 21: If $\text{Var}X$ exists then

$$\text{Var}(aX + b) = a^2 \text{Var}X.$$

Examples

Example 45: ($X \sim U[a, b]$)

Then $E(X) = \frac{a+b}{2}$ and $E(X^2) = \frac{1}{3}(b^2 + ab + a^2)$. Hence

$$\text{Var}X = \frac{1}{12}(b - a)^2.$$

Example 46: ($X \sim \text{BIN}(n, p)$)

Then $E(X) = np$ and $E(X^2) = n(n-1)p^2 + np$. Hence

$$\text{Var}X = n(n-1)p^2 + np - n^2p^2 = np - np^2 = np(1-p).$$

Interpretation of variance

- Discrete case: $VarX = \sum_i (x_i - E(X))^2 P(X = x_i)$. The variance is small if each term in the sum is small. This when the x_i 's with large differences $|x_i - E(X)|$ have low probabilities. In other words, when the values of X are closely concentrated around $E(X)$.

Theorem 22: The mean square deviation from ξ (i.e. $E[(X - \xi)^2]$) is minimized at $\xi = E(X)$ and its minimal value equals $VarX$.

Theorem 23: The mean absolute deviation from ξ (i.e. $E|X - \xi|$) is minimized at $\xi = m$, where m is median of X .

Skewness and kurtosis

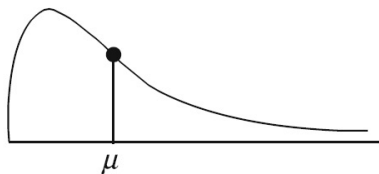
Definition 20: Let X be a random variable with $E(X) = \mu$, $VarX = \sigma^2 > 0$ and such that the third central moment $\gamma_3 = E[(X - \mu)^3]$ exists. The ratio $\frac{\gamma_3}{\sigma^3}$ is called the **coefficient of skewness**.

Definition 21: Let X be a random variable with $E(X) = \mu$, $VarX = \sigma^2 > 0$ and such that $\gamma_4 = E[(X - \mu)^4]$ exists. The ratio $\frac{\gamma_4}{\sigma^4}$ is called the **coefficient of kurtosis**.

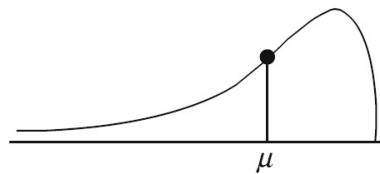
Example 47: Find skewness and kurtosis of the following distributions:

- i) $U[0, 1]$,
- ii) $BIN(1, p)$.

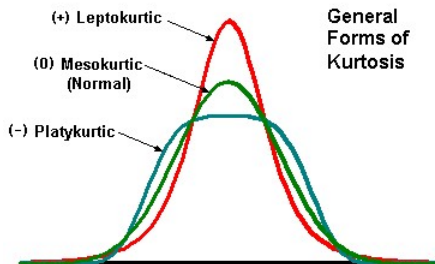
Skewness and kurtosis



$\mu_3 > 0$, skewed to the right



$\mu_3 < 0$, skewed to the left



Chebyshev inequality

Theorem 24: (Chebyshev's inequality)

For any random variable X with $\text{Var}X = \sigma^2$, $0 < \sigma < \infty$, for every $\varepsilon > 0$

$$P(|X - E(X)| \geq \varepsilon) \leq \frac{\sigma^2}{\varepsilon^2}.$$

Example 48: We toss a coin 20 times and want to estimate the probability that the number of heads deviates from 10 by 3 or more.

$$E(X) = np = 10, \text{Var}X = \sigma^2 = np(1 - p) = 5.$$

Using Chebyshev's inequality, we find

$$P(|X - 10| \geq 3) \leq \frac{5}{9} \approx 0.5555.$$

However, the exact probability equals [0.2632!](#)

- While Chebyshev's inequality is not very impressive in many cases (such as Example 48), it remains valid for all random variables.

Beyond Chebyshev's inequality

- Among the most important applications of Chebyshev's inequality are the laws of large numbers (see Chapter on Random Samples).

Example 49: Consider the binomial rv S_n (the number of successes in n independent trials). If p is the probability of success, and $q = 1 - p$, then

$$E(S_n/n) = p, \quad \text{Var}(S_n/n) = pq/n.$$

Using Chebychev's inequality, we obtain

$$P\left(\left|\frac{S_n}{n} - p\right| > \varepsilon\right) \leq \frac{pq}{n\varepsilon^2}.$$

For every $\varepsilon > 0$, as $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} P\left(\left|\frac{S_n}{n} - p\right| > \varepsilon\right) = 0.$$

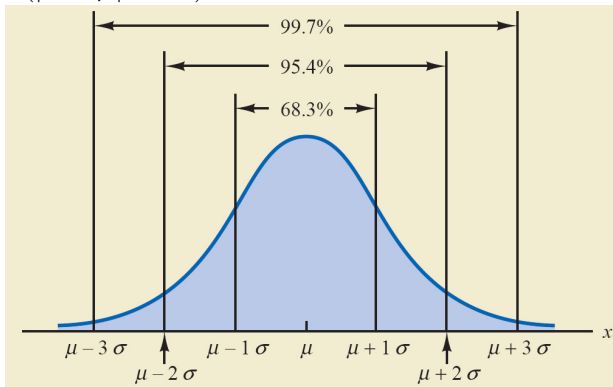
This explains why the empirical frequency of an event approaches the probability of the event as number of trials increases.

3- σ rule for normal distribution

- Consider $Z \sim N(0, 1)$ then

$$P(|Z| > z) = 2(1 - \Phi(z)).$$

- since $\Phi(3) = 0.9987$ then for $X \sim N(\mu, \sigma^2)$ we have $P(|X - \mu| > 3\sigma) = 0.0026$.



Covariance

Consider random variables X and Y . Then

$$\begin{aligned}\text{Var}(X + Y) &= E(X + Y)^2 - (E(X + Y))^2 = \\ &= E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2 = \\ &= E(X^2) - (E(X))^2 + E(Y^2) - (E(Y))^2 + \\ &\quad + 2(E(XY) - E(X)E(Y))\end{aligned}$$

Definition 22: The quantity

$$\text{Cov}(X, Y) = E(XY) - E(X) \cdot E(Y)$$

is called the **covariance** of random variables X and Y , provided $E(X^2) < \infty$ and $E(Y^2) < \infty$ (i.e., $E|XY| < \infty$).

Example 50:

$$\text{Cov}(X, X) = E(X^2) - (E(X))^2 = \text{Var}X.$$

Variance and covariance of linear combinations of rv's

Theorem 25: If $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ are two sets of random variables with finite second moments then

$$\text{Cov} \left(\sum_{i=1}^n a_i X_i + c, \sum_{j=1}^m b_j Y_j + d \right) = \sum_i \sum_j a_i b_j \text{Cov}(X_i, Y_j).$$

Theorem 26: Let $X_1, \dots, X_n$ be random variables with $E(X_i^2) < \infty, i = 1, \dots, n$. Then

$$\text{Var}(a_1 X_1 + \dots + a_n X_n + b) = \sum_{j=1}^n a_j^2 \text{Var} X_j + 2 \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j).$$

Example 51: Let $a_1 = 1, a_2 = -1$ and $b = 0$. Then

$$\text{Var}(X - Y) = \text{Var} X + \text{Var} Y - 2\text{Cov}(X, Y).$$

Independent vs. uncorrelated rv's

Definition 23: We call rv's X and Y **uncorrelated** if

$$\text{Cov}(X, Y) = 0.$$

- covariance can be understood as a signed measure of the variation of X relative to Y .
- uncorrelated random variables X and Y indicate that there is linear dependence between them. However, this does not imply that X and Y are independent random variables!

Theorem 27: Let $X_1, \dots, X_n$ be pairwise uncorrelated random variables with $E(X_i^2) < \infty, i = 1, \dots, n$. Then

$$\text{Var}(a_1X_1 + \dots a_nX_n + b) = \sum_{j=1}^n a_j^2 \text{Var}X_j.$$

In particular, it holds when $X_1, \dots, X_n$ are independent.

Averaging of random variables

Example 52: If $X_1, \dots, X_n$ are independent random variables with the same distribution, then their average

$$\bar{X}_n = \frac{X_1 + \dots + X_n}{n}$$

has variance given by

$$\text{Var}\bar{X}_n = \frac{\text{Var}X_1}{n}.$$

Thus, averaging reduces variability (in terms of variance) by a factor of $1/n$.

Foot as a measure: 0.3048 meters exactly (as of 1960)

Example 53: (foot as a measure)

Effect of averaging on variability was understood long before it was mathematically explained. For instance, there was a 16th century law in Frankfurt that defined the length of one foot.



Stand at the door of a church on a Sunday and bid 16 men to stop, tall ones and small ones, as they happen to pass out when the service is finished then make them put their left feet one behind the other, and the length thus obtained shall be a right and lawful rood to measure and survey the land with, and the 16th part of it shall be the right and lawful foot.

- i) How much this procedure cuts down the variability (measured by the standard deviation) of the length of feet?
- ii) Assume that a man's shoe has an average length of 1 foot and $\sigma = 0.1$ foot. Find the (approximate) probability that the mean of 16 lengths of men's shoes exceed 1 foot by more than 1 inch. Assume normality.

Correlation coefficient

Definition 24: Let X and Y be two random variables with finite second moments. Then for $\text{Var}X > 0$ and $\text{Var}Y > 0$

$$\text{Corr}(X, Y) = \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}X \cdot \text{Var}Y}}$$

is called the **coefficient of correlation** between X and Y .

Example 54: Let X and Y be discrete random variables with the following joint values $(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)$, each with a probability $1/5$.

Then $E(X) = 0, E(Y) = 2, E(XY) = 0, \text{Cov}(X, Y) = 0, \rho_{X,Y} = 0$.

Thus, X and Y are not independent, yet they are uncorrelated.

Example 55: Let X and Y be continuous random variables with a joint uniform distribution on the square defined by the vertices $(-1, 0), (0, -1), (1, 0)$ and $(0, 1)$.

Then $E(X) = 0, E(Y) = 0, E(XY) = 0, \text{Cov}(X, Y) = 0, \rho_{X,Y} = 0$.

Thus, X and Y are not independent, even though they are uncorrelated.

Schwarz inequality and properties of $\rho_{X,Y}$

Theorem 28: (Schwarz inequality)

For any random variable X and Y

$$(E(XY))^2 \leq E(X^2)E(Y^2).$$

Theorem 29: The coefficient of correlation $\rho_{X,Y}$ satisfies

a)

$$|\rho_{X,Y}| \leq 1$$

with $|\rho_{X,Y}| = 1$ if and only if there is a constant $a \neq 0$ and b such that $P(Y = aX + b) = 1$.

b) for any a, b, c, d with $ac \neq 0$

$$\rho_{aX+b, cY+d} = \begin{cases} \rho_{X,Y}, & \text{if } ac > 0; \\ -\rho_{X,Y}, & \text{if } ac < 0. \end{cases}$$

Recommended reading (week 4)

For the next lecture Part 5, please, read

- sections 6.3, 7.7 from Bartoszynski and Niewiadomska-Bugaj (11 pages)
- sections 2.6, 3.5 from Mittelhammer (12 pages)